

Make training work

AICE3002 / AICE6001

VLC = $\int \int \int$ Vision
Learning α VoLaC
Control

Practical Training of Deep Networks

Normalisation, stabilisation, and diagnosis

Jonathon Hare

Vision, Learning and Control
University of Southampton

Training is a coupled dynamical system

Initialisation recap: Xavier, He, and PyTorch defaults

Normalisation layers control activation scale

Batch normalisation

BatchNorm during training and evaluation

Batch size and data dependence

Initialisation and normalisation complement each other

Gradient clipping limits unstable updates

Learning-rate schedules and warm-up

Weight decay connects optimisation and regularisation

Training curves give diagnostic evidence

Inspecting activations and gradients

Dead and saturated units

A systematic failure-diagnosis workflow